

proxy_forward_benchmark.sh — user guide

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Overview

tests/benchmark/proxy_forward_benchmark.sh is the §11.4.169 **benchmarking / performance** test-type for the live HTTP forward proxy (Squid on 127.0.0.1:53128). It measures forward-proxy request **latency** (nearest-rank p50 / p95 / p99 + min / max / mean) and sequential **throughput** (successful requests per wall-second) by driving N (default 200) plain-HTTP requests through the proxy to a 204-returning endpoint and capturing **every** request's `%{http_code} + %{time_total}` to an evidence file. The verdict rests on the captured per-request samples, never a summary number (§11.4.6 no fake numbers, §11.4.69 captured evidence).

It is the performance sibling of the §11.4.85 stress suite (tests/stress/proxy_forward_stress.sh) and reuses the same anti-bluff verdict contract from tests/lib/evidence.sh.

Prerequisites

- The base proxy stack UP (./start), Squid listening on :53128.
- bash, curl, awk, sort, grep on PATH.
- Network reachability to BENCH_TARGET (default www.gstatic.com) — an external outage is classified as an honest SKIP, never a false FAIL.

Usage examples

```
# default: 200 sequential requests, host-safety caps applied
GOMAXPROCS=2 nice -n 19 ionice -c 3 bash tests/benchmark/
proxy_forward_benchmark.sh
```

```
# smaller/larger sample, custom proxy/target
BENCH_N=50 PROXY_ADDR=127.0.0.1:53128 \
```

```
BENCH_TARGET=http://www.gstatic.com/generate_204 \  
bash tests/benchmark/proxy_forward_benchmark.sh
```

Env overrides: PROXY_ADDR (default 127.0.0.1:53128), BENCH_TARGET, BENCH_EXPECT (default 204), BENCH_N (default 200), CURL_MAX_TIME (default 20), BENCH_EVIDENCE_DIR.

Edge cases

- **Proxy dropped requests while target reachable directly** → OVERALL=FAIL (exit 1) — a real proxy defect (§11.4.68 no fail-open, never masked as SKIP).
- **Target unreachable via proxy AND directly, port listening** → SKIP:network_unreachable_external (exit 3) — third-party outage, not a proxy defect (§11.4.1 no false-FAIL of a healthy proxy).
- **Proxy port not listening, no direct signal** → SKIP:topology_unsupported (exit 3) — the base stack is down (§11.4.3).
- A hung request cannot exceed CURL_MAX_TIME (bounds the whole run) and is recorded as 000.

Internal behaviour

1. Sources tests/lib/evidence.sh (walks up to find the repo root).
2. Fires N sequential curl -x http://\$PROXY_ADDR \$BENCH_TARGET, appending %{time_total} to a sample set and counting EXPECT responses.
3. Sorts the samples, computes nearest-rank percentiles + throughput, writes latency.txt (raw samples + distribution) and benchmark.evidence under qa-results/benchmark/proxy_forward_<ts>/.
4. Cross-checks a direct (non-proxied) probe to classify a shortfall as a proxy defect vs an external outage.
5. Emits one structured PASS ($\geq$ N successful 204s) / FAIL / SKIP verdict via ab_pass_with_evidence / ab_skip_with_reason, citing latency.txt.
6. trap cleanup removes only the scratch dir (§11.4.14); evidence preserved.

Last verified

2026-07-01 — live run against the base proxy: **200/200** proxied 204s, p50=0.086s p95=0.088s p99=0.091s (min 0.085 / max 0.102 / mean 0.087), throughput 10.841 req/s, OVERALL=PASS. Evidence: qa-results/benchmark/proxy_forward_20260701T130414Z/latency.txt.

Related scripts

- `tests/stress/proxy_forward_stress.sh` — §11.4.85 stress sibling.
- `tests/lib/evidence.sh` — shared anti-bluff verdict helpers.
- `docs/design/hardening/Status.md` — §11.4.169 test-type coverage matrix.